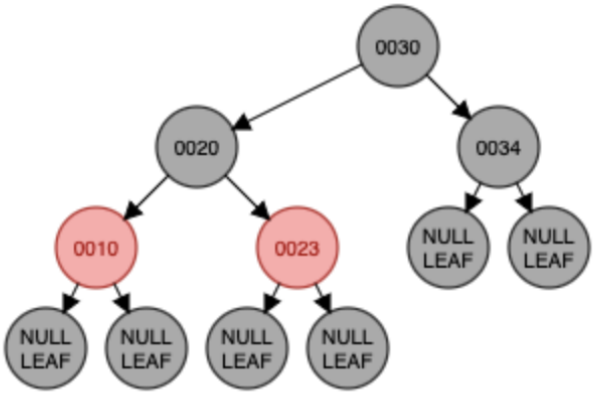


1- After inserting the node with key 25 into the Red-Black Tree shown above, select all the correct statements:
(a) A left rotation over the node with key 23 is required to maintain the properties of the Red-Black Tree.
(b) The nodes 20 and 25 will be the only red nodes in the tree to preserve its Red-Black properties.
(c) The height of the BST is 3 (excluding the null leaves).
(d) The in-order traversal of the tree is 20, 10, 34, 23, 25, 30.

2 - When a new activity is started in Android, what happens to the previous activity in the activity stack?

(a) The previous activity is destroyed immediately; onPause() → onStop() → onDestroy() are called.

(b) The previous activity is pushed to the back stack and paused.

(c) The previous activity remains active in the stack and no lifecycle methods are called.

(d) The previous activity is removed from the stack but kept in memory; only onStop() is called.

3 - *Considering the grammar below and assuming that there is a parser for it, which instruction is NOT acceptable for this grammar?*

(a) fetch name, age in Person having id = -12
(b) fetch street in Address having id = 1
(c) fetch name, age, height in Person having id 2
(d) fetch name in Person having id = 4

<command> := fetch <fields> in <table_name> having id = <id_value>
<fields> := <field>, <fields> | <field>
<table_name> := <string_value>
<id_value> := <int_value>
<field> := <string_value>

4 - Which of the following **cannot** be achieved using JML (Java Modeling Language)?
(a) Formally specify preconditions, postconditions, and invariants for Java methods and classes.
(b) Perform runtime assertion checking.
(c) Document detailed design decisions.
(d) Support design by contract.
(e) Handle Java XML processing.

5 - *Considering the class below, which alternative is CORRECT:*

a. *The JUnit assertion assertSame(Universe.getInstance(), Universe.getInstance()) will fail.*
b. *The JUnit assertion assertEquals(Universe.getInstance(), Universe.getInstance()) will fail.*
c. *You cannot instantiate a Universe instance directly using the constructor outside the class.*
d. *You can remove the static modifier from the field INSTANCE and from the method getInstance() if you want to implement a Singleton.*

```
public class Universe {

    private static Universe INSTANCE;

    private Universe() { }

    public static Universe getInstance() {
        if(INSTANCE == null) {
            INSTANCE = new Universe();
        }
        return INSTANCE;
    }
}
```